# Every Wave Is Three Numbers And A Phase

**SYNOPSIS**

A wave in physics is an umbrella term encompassing all things repetitive. When physicists hear the word “wave,” they think of oscillations, repeating patterns, cyclical behaviours — things that extend or spread over space and time. They also think of the sinusoidal functions (sines and cosines) that can describe all such repetitions. And a sinusoid needs only three numbers to fix its shape: **amplitude**, **wavelength** and **frequency** — plus one more idea, **phase**, the moment you want to compare two waves. You have already met the three numbers and watched what each one does to a wave, and you will see before the end that the medium ties the last two of them together. In this note we pull them together, put them on a graph, and use them to work out one more thing the first two notes could not give you: how fast the wave actually travels.

**You will need:** this handout, a pencil, a ruler and a calculator. Every number you need is printed on Figs. 1 and 2 — no equipment, no simulation. Write on the figures; that is what they are for.

## Waves carry energy

Think about the energy in waves. If you see massive ocean waves crashing into shores, toppling buildings, trees and other structures, clearly there is an enormous amount of energy in those waves. That energy came from somewhere. Perhaps there was an undersea earthquake that generated a tsunami — so the energy in those ocean waves started as potential energy in locked-up tectonic plates, under tremendous tension and then abruptly released. The earthquake could have happened thousands of miles from the shore and still inflict massive damage, *because the waves carry the energy*. The water that does the waving serves as the **medium** that allows the energy to be transmitted. An ocean wave is an example of a **mechanical wave**: it can carry energy as long as there is an expanse of water that can “wave.” Many wave phenomena are mechanical waves and require a medium to propagate through. Waves on a string need the string as the medium to carry the energy of the shaker, as you saw in the PhET simulation in Note 2.

Not every wave works this way, and the exception is worth naming now because it will keep turning up. Light, radio, microwaves and x-rays are **electromagnetic waves**, and they need no medium at all — sunlight crosses 150 million kilometres of empty space to reach us. The classic demonstration is a ringing bell inside a jar: pump the air out and you can still *see* the bell shaking, but you can no longer *hear* it. Take away the medium and the mechanical wave dies; the electromagnetic one carries on. Everything else in this course is mechanical, so “needs a medium” is the safe assumption — but sound is mechanical and light is not, and that one line separates them.

## What actually moves

Now that you see that waves carry energy, what does the mechanical motion of the medium look like? In other words, what motion carries the energy? Take ocean waves again, with water as the medium. If you could “chunk” sections of the ocean, the wave is nothing more than these chunks of water moving up and down individually — there is no *flow* of ocean water from the earthquake to the shore. A perfect analogy is the “Mexican wave” you sometimes see in a stadium when the spectators are bored. The key feature of that wave is that it sweeps through and goes right around the stadium without halting — and crucially, the spectators only stand up and sit back down. They never shift from their seats. They only need to stand and sit at the right moment, and that timing is what dictates the wavelength and frequency of the wave. So the medium transports the energy, but not the material the medium is made of.

## Two ways for the medium to move

Ocean waves, sound waves and waves on a string are all mechanical waves in which the mechanical motion of the medium transmits the energy of the wave generator, while the material the medium is made of is not transported. Ocean waves and waves on a string are just like the Mexican wave: the wave moves, let us say, from left to right, but the spectators themselves move up and down. Here the direction of wave travel ("left to right") is *perpendicular* to the medium's mechanical motion ("up and down"). This type of wave is called a **transverse wave** (*trans*— across).

A sound wave, in contrast, is a repeating pattern of high and low air pressure moving through space. The high and low pressure is the bunching up of air molecules (high pressure) and the spacing out of air molecules (low pressure), caused by a chunk of air repeatedly moving "left–right, left–right." The pressure pattern itself, meanwhile, travels from left to right. In other words, the direction of wave travel is *parallel* to the mechanical motion of the medium. This type of wave is called a **longitudinal wave** (*longitudinal* — along).

## Phase

The last idea is the **phase** of a wave. When you hear "phase" in physics, picture a *moon phase*. A phase of the moon is a specific part — a specific place — in the cycle of moon shapes: the full moon phase is the spot in the cycle where the moon is a complete disc. A phase of a wave is a specific place in the wave, and that is all it is. You can say peak phase, or trough phase. And once you have the word, you can define the other terms with it: the **wavelength** is the distance between identical phases in adjacent cycles, and the **period** is the duration between identical phases in adjacent cycles. For example, the wavelength is the distance from one peak phase to the peak phase of the next cycle.

Phase is most useful when you are comparing two waves. Imagine two waves travelling in the same medium with the same wavelength (so, at the same speed, the same frequency too — why?), and lay one on top of the other. You might find that all the phases align: the peak phase of one sits on the peak phase of the other. That is **in phase**. Or you might find the peak phase of one lining up with the trough phase of the other — one wave is *shifted in phase* with respect to the other. Peak on trough is **180° out of phase**, or completely out of phase. When the phases of two waves do not align at all, we simply say the waves are out of phase.

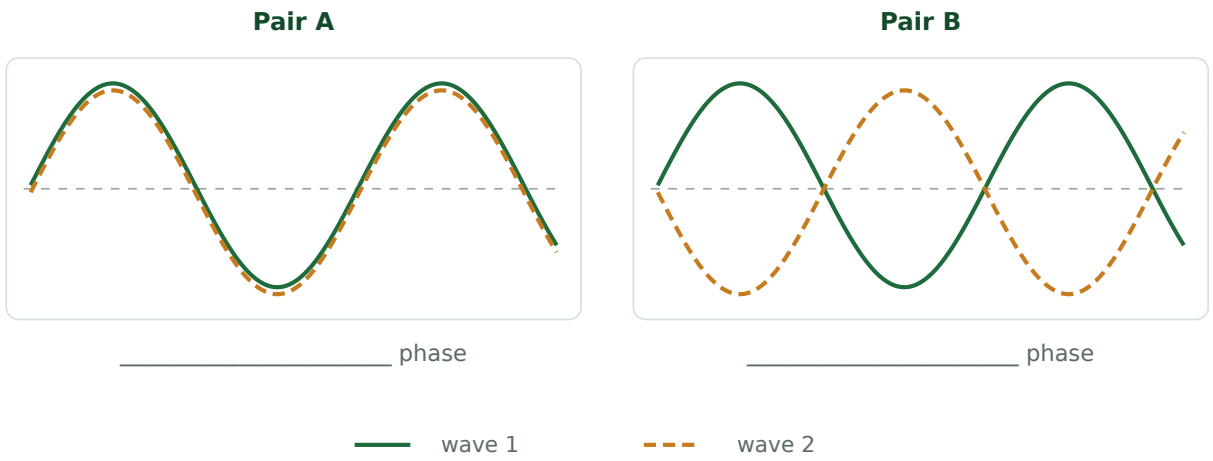



**Fig 3.** Two pairs of waves of equal wavelength, laid one on top of the other. In each pair the two waves are drawn a hair apart vertically so you can see both — read only the horizontal alignment of their peaks and troughs. Question 14 asks you to fill in the two blanks.

## Performance Expectations

Explain amplitude, wavelength and frequency in waves. Describe the velocity of a wave using equations in terms of wavelength and frequency. Describe mechanical waves, and how a mechanical wave carries energy but not material. Describe, compare and contrast transverse and longitudinal waves, and give examples of each. Use the phase of a wave to describe wavelength and period.

## Instructions

In this note you will summarise the important terms in waves using diagrams and equations, then answer the questions that follow. Figs. 1 and 2 are two views of *the same wave*. Read your numbers straight off the grid — every number you need is on one of the two figures. Write your answers in the space provided.

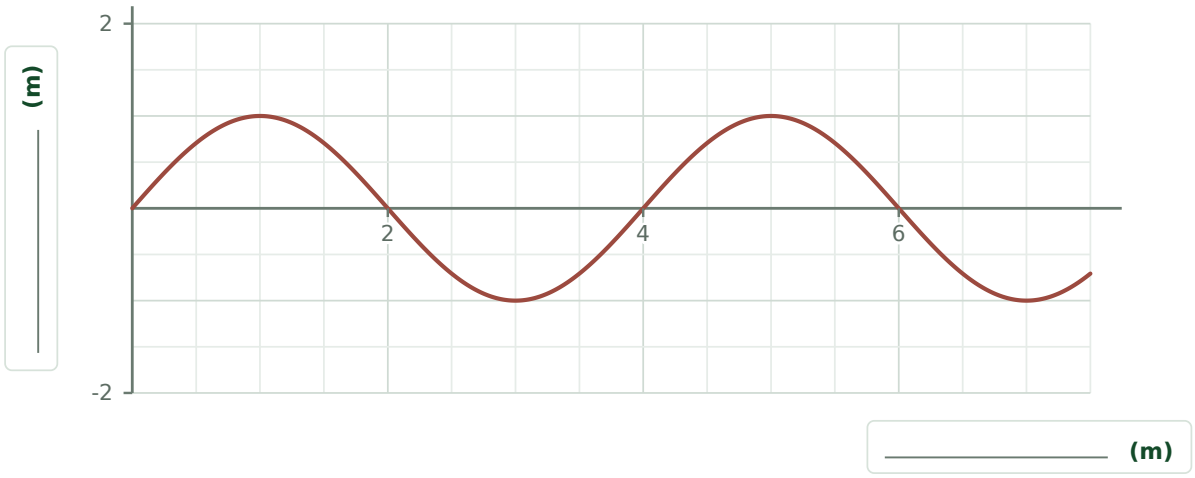



**Fig 1.** The wave plotted against **distance**: a snapshot, frozen at one instant, of the whole medium at once. Both axis names have been left blank on purpose — that is question 2. The unit is given so you can still read numbers off the grid. Fine gridlines are every 0.5 units.

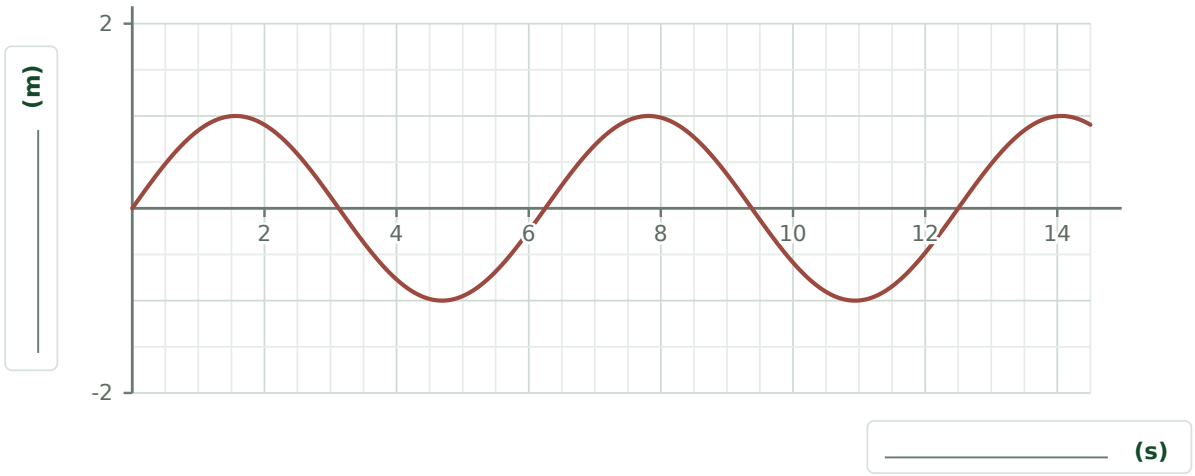



**Fig 2.** The **same wave** plotted against **time**: what one fixed spot in the medium does as the wave goes past, as if you stood still and measured the height in front of you. Again the axis names are yours to fill in.

**1** Figs. 1 and 2 represent a moving wave — the *same* wave. One of them is a snapshot of the whole medium frozen at a fixed instant; the other is the history of one fixed position as time passes. Say which figure is which, and state which one you would use to find a *wavelength* and which to find a *period*.

**2** **Label the axes** in Figs. 1 and 2. Write the *name of the quantity*, not just the unit, on each of the four blanks.

**3** **Read** the amplitude from the figure.

**Amplitude, A =** _____ m

**4** **Indicate** the amplitude on the figures — do it on both.

✎ **Mark this directly on the figure.**

**5** Determine the **wavelength** from Fig. 1, and **indicate** it on the figure.

✎ **Mark this directly on the figure.**

**Wavelength, λ =** _____ m

**6** Determine the **period** from Fig. 2, and **indicate** it on the figure.

✎ **Mark this directly on the figure.**

**Period, T =** _____ s

**7** What is the **frequency** of this wave? Show the unit.

**Frequency, f =** _____

**8** Imagine standing still — in one fixed position — on a beach, watching the waves come in and lap at your feet. How would you determine the **speed** at which those waves are arriving? Describe the method.

**9** Determine the **velocity** of the wave represented in Figs. 1 and 2. **State the equation first**, then put your numbers in.

**v =** _____ (equation)

**v =** _____ (show the unit)

**10** A tsunami crosses an entire ocean and wrecks a shoreline thousands of miles from the earthquake that started it. Yet the water lapping at that shore did *not* come from the earthquake. Say what made the trip and what did not — and say what the water is doing while all this happens.

**11** What is a **transverse** wave? Draw one, and give examples.

**A — Transverse**

Draw the wave, then draw and label two arrows: how the medium moves, and which way the wave travels. They should end up at right angles.

**12** What is a **longitudinal** wave? Draw one, and give examples.

**B — Longitudinal**

Draw the wave, then draw and label the same two arrows. This time they should end up along the same line.

**13** What are the **similarities and differences** between transverse and longitudinal waves?

**Similarities**

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**Differences**

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**14** What is a **phase** in waves? Then look at Fig. 3 and fill in the two blanks under the panels: one pair is *in phase*, the other is *180° out of phase*. Say how you can tell.

✎ **Mark this directly on the figure.**

**15** Use the phase to describe the velocity of a wave. Complete the sentence — or come up with your own phrase that uses the word “phase.”

**“The speed of a wave is the speed of a c _____ (8-letter word) phase.”**

**WHERE THIS IS GOING**

You now have every number a single wave can have: A, λ, f, T, and the speed that ties them together. Notice what the last section quietly added — a way to talk about *two* waves at once. Once two waves are in the same place, their phases decide everything about what you see and hear: peaks on peaks reinforce, peaks on troughs cancel. That is interference, and it is where the rest of this course lives.

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